

HDOC - Day 13 Activity

C Programming Assignment

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits. E.g. Number: 12045, Even_dig = 3, Odd_dig = 2
- Find the sum of its even digits and the sum of its odd digits separately. E.g. Number: 12345, Sum_even = 6, Sum_odd = 9

Algorithm

1. START
2. Input a positive integer n.
3. If $n \leq 0$, display an error message and STOP.
4. Display the menu:
 - 1 - Count even and odd digits
 - 2 - Find sum of even and odd digits
5. Input choice.
6. Set $temp = n$.
7. Use switch(choice).
8. If choice = 1:
even_count = 0, odd_count = 0
WHILE $temp > 0$:
digit = $temp \% 10$

```

IF digit % 2 == 0:
even_count = even_count + 1
ELSE:
odd_count = odd_count + 1
temp = temp / 10
Display even_count and odd_count.
9. If choice = 2:
sum_even = 0, sum_odd = 0
WHILE temp > 0:
digit = temp % 10
IF digit % 2 == 0:
sum_even = sum_even + digit
ELSE:
sum_odd = sum_odd + digit
temp = temp / 10
Display sum_even and sum_odd.
10. Otherwise, display "Invalid choice".
11. STOP

```

Test Case Table

Test	Input(n)	Choice	Expected Output	Result
1	12045	1	Even digits= 3 , Odd digits= 2	Pass
2	12345	2	Sum even= 6 , Sum odd= 9	Pass
3	24680	1	Even digits= 5 , Odd digits= 0	Pass
4	13579	2	Sum even= 0 , Sum odd= 25	Pass
5	12345	3	Invalid choice	Pass
6	-25	1	Enter a positive integer only.	Pass

C Code

```

#include <stdio.h>
int main() {
int n, temp, digit, choice;
int even_count = 0, odd_count = 0;
int sum_even = 0, sum_odd = 0;
printf("Enter a positive integer: ");

```

```
scanf("%d", &n);
if (n <= 0) {
printf("Enter a positive integer only.\n");
return 0;
}
printf("\nMENU\n");
printf("1. Count even and odd digits\n");
printf("2. Find sum of even and odd digits\n");
printf("Enter your choice: ");
scanf("%d", &choice);
temp = n;
switch (choice) {
case 1:
while (temp > 0) {
digit = temp % 10;
if (digit % 2 == 0)
even_count++;
else
odd_count++;
temp = temp / 10;
}
printf("Even digits = %d\n", even_count);
printf("Odd digits = %d\n", odd_count);
break;
case 2:
while (temp > 0) {
digit = temp % 10;
if (digit % 2 == 0)
sum_even += digit;
else
sum_odd += digit;
temp = temp / 10;
}
printf("Sum even = %d\n", sum_even);
printf("Sum odd = %d\n", sum_odd);
```

```

break;
default:
printf("Invalid choice\n");
}
return 0;
}

```

Compilation , Execution and Testing Cases

```

Session Actions Edit View Help
(kali@kali)-[~]
$ vi evenodd.c
(kali@kali)-[~]
$ cc evenodd.c
(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12045

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 3
Odd digits = 2

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum even = 6
Sum odd = 9

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 24680

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 5
Odd digits = 0

```

```

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 13579

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even digits = 0
Odd digits = 5

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 13579

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum even = 0
Sum odd = 25

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 3
Invalid choice

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: -25
Enter a positive integer only.

```